

Engineering Portfolio



Humza Khan (B.S. Aerospace Engineering)

hhk.hamza123918@gmail.com

516-225-4887

Humza Khan | LinkedIn

Website (Access Documents Here!)



UB AIAA – Design/Build/Fly (3 years)

Leadership/Mentorship

- Held multiple leadership roles including **CNC Laser Cutting Specialist**, **Events Coordinator**, and **Vice President** of the club.
- Led multiple workshops and training sessions teaching new members about safe operations of lab tools and equipment.
- Managed club budget (~\$40,000) to source build material and plan competition trips/logistics for 23 active team members.
- Sourced external funding by networking with industry professionals in Buffalo area and school faculty.

Accomplishments & Results

- Used tools like **ANSYS Fluent** and **XFLR5** to predict the aerodynamic performance of multiple aircraft prototypes.
- Used CAD tools like **SOLIDWORKS** and **Fusion360** to model aircraft designs and performed FEA via **ANSYS Static Structural** to predict the structural performance when the designed model is under load.
- Manufactured critical aircraft components using **carbon fiber** and **epoxy** on **3D printed molds** in **vacuum bags**.
- Manufactured critical structural components like motor mounts via **CNC Laser cutting** and aerodynamic components like wings via **CNC Hot Wire cutting**.
- Placed **12th** out of 112 teams internationally that participated in DBF 2025. A huge improvement compared to 53rd when I first joined the organization.



Senior Design Project (Hermes)

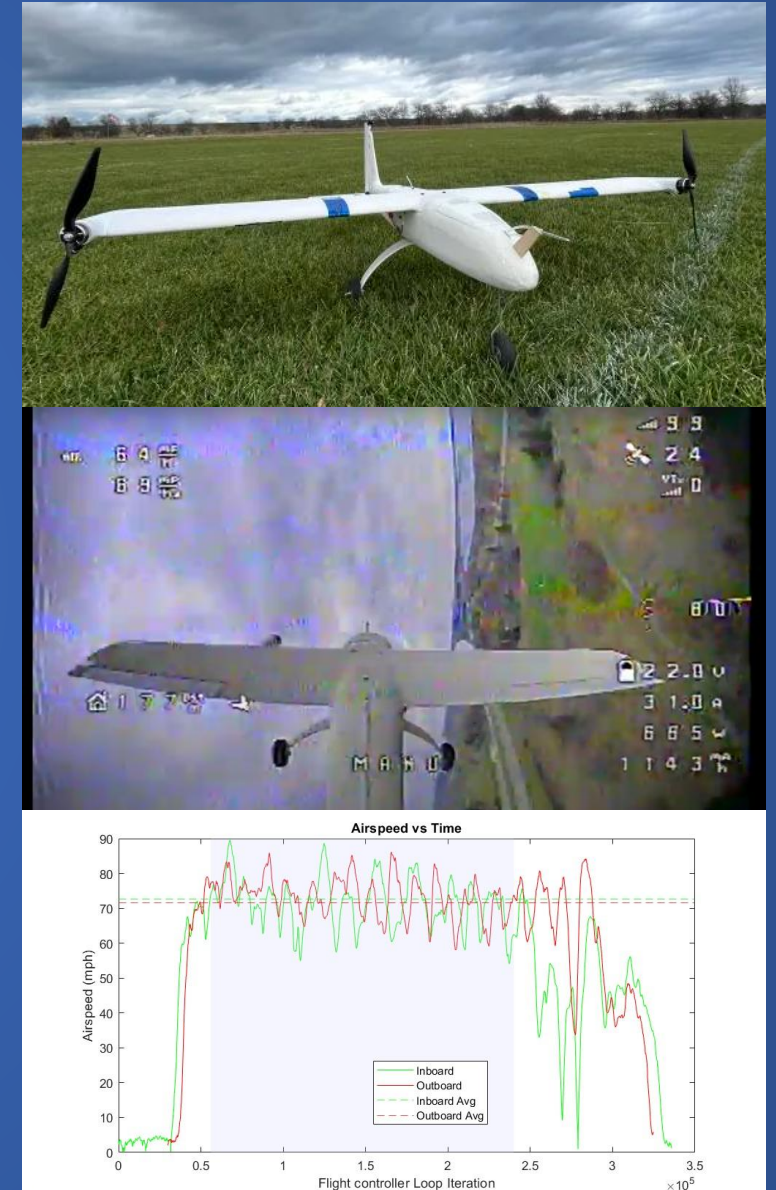
(Wingtip Motor Study)

Objective

- Determine the effect of wingtip mounted motor propulsion as compared to mid-wing mounted motors.
- Some studies indicated that wingtip mounted propellers can significantly reduce induced drag by minimizing the generation of wingtip vortices.

Methods & Results

- Designed and manufactured a 7lb, 5ft wingspan UAV with modular motor attachment mounts on the wings.
- The UAV was constructed using CNC hotwire-cut foam cores, fiberglass layups, and 3D-printed components. Final shaping and assembly involved hand tools like sanding blocks and foam saws.
- Conducted multiple successful test flights where the data was stored on an onboard SD card. The data was later processed through MATLAB.
- Flight data logged via onboard SD card showed consistent performance across both motor configurations, with a peak airspeed of ~107 mph. MATLAB post-processing enabled comparison of stability and control responsiveness.



Aircraft Design Project (Latos)

(Range Optimization and STOL Performance)

Objective

- Design a **Short Takeoff and Landing (STOL)** capable business jet to serve **unprepared airstrips** and **remote airports**.
- Target performance included:
 - **3000 NM range**
 - **1200 ft takeoff/landing field**
 - Compliance with **FAR 25** certification criteria.
- Emphasis on blending **utility aircraft STOL performance** (e.g., DHC-6 Twin Otter) with **long-range business jet capabilities** (e.g., Gulfstream G650).

Methods & Results

- My main role was to write **MATLAB** scripts that converges design parameters to the set requirements/objectives.
- Selected **T-tail, over-wing engine** configuration for blown flap compatibility and better clearance in rugged conditions.
- Integrated **triple-slotted flaps** and **leading-edge slats** for enhanced lift at low speeds.
- Final optimized configuration achieved:
 - **Takeoff distance ≈ 1035 ft**
 - **Cruise at Mach 0.53 (~ 350 mph)**
 - **Fuel stored in self-sealing rubber bladders** inside wings.
- Structural design featured **aluminum longerons**, **high-wing configuration**, and **tundra tire landing gear** in fuselage pods.



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Initial Velocity Calculations :)
Cruise Velocity: 515.00 ft/s
Speed of Sound at 40,000 ft: 968.12 ft/s
Cruise Mach Number: 0.53
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Engine Aircraft/Aerodynamics/Payload Specifications :)
Payload Weight: 3330.00 lb
Crew Weight: 370.00 lb
Aspect Ratio: 8.00
Reference Surface Area Ratio: 4.34
-----
Range & Loiter Parameters :)
Range: 18228000.00 ft
Loiter time: 2700.00 seconds
Endurance: 35394.17 seconds
-----
Engine Properties - High-bypass turbofan :)
C_loiter: 1.111111e-04
C_cruise: 1.388889e-04
-----
L/D Calculation :)
L/D (MAX): 18.56
L/D (CRUISE): 16.07
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3D Printed Glider Project

(Aero-Lab Class Competition)

Objective

- Design and fabricate a 3D-printed glider to achieve maximum glide distance using only 3D-printed materials, glue, and paint.
- Ensure aerodynamic performance and structural integrity suitable for competition launch and landing.

Methods & Results

- **Precision Manufacturing:** Fabricated critical airframe components using Brass HTPLA for nose weight balancing and LW-PLA for lightweight aerodynamic structures.
- **Material Optimization:** Applied epoxy coatings and reinforcement stringers to improve strength without sacrificing weight.
- **Printing Techniques:** Calibrated Bambu Lab A1 printer settings (temperature, flow rate, infill) for consistent surface finish and dimensional accuracy.
- **Flight Testing & Iteration:** Early designs suffered from tail-heaviness and structural failures. Revised prototypes improved center of gravity and coating quality using Aeropoxy, resulting in smoother flight and better stability.
- **Final Outcome:** The glider achieved stable flight but failed on landing, offering key lessons in tail reinforcement, CG adjustment, and iterative prototyping.



Experimental Aerodynamics & Heat Transfer

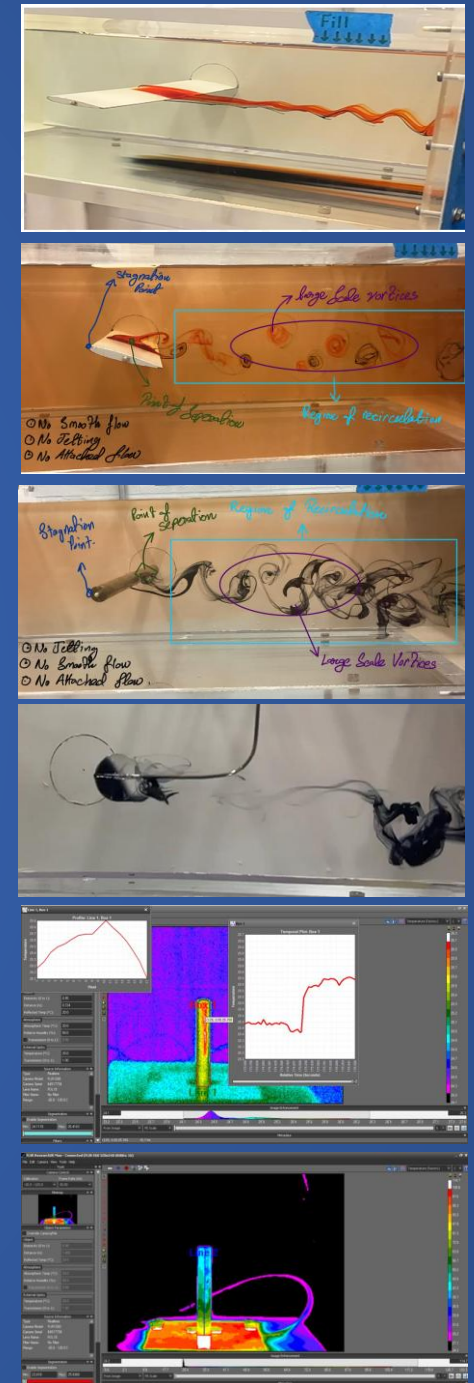
(MAE339 Aero-Lab Experiments)

Objective

- Experimentally investigate fluid mechanics and heat transfer phenomena.
- Validate theoretical aerodynamic and thermal models using measured data.
- Perform uncertainty analysis and compare steady vs unsteady flow behavior.
- Process and interpret experimental data using MATLAB and regression tools.

Methods & Results

- Conducted flow visualization, wind tunnel, heat transfer, and unsteady drag experiments to study boundary layer behavior, aerodynamic forces, and convection.
- Observed flow separation near stall ($\sim 8\text{--}14^\circ$ AoA), vortex shedding behind bluff bodies, and Reynolds-number-dependent wake behavior.
- Measured pressure distributions on NACA 4412 airfoil; computed C_l , C_d , C_m , and center of pressure using MATLAB numerical integration.
- Wind tunnel data showed stall near $8\text{--}9^\circ$ AoA and negative C_m slope, confirming longitudinal stability and agreement with published airfoil data.
- Analyzed cylindrical fin under forced convection; calculated fin efficiency ≈ 0.93 and heat transfer rate ≈ 44.6 W, validating 1D fin theory.
- Determined drag coefficient using unsteady (full-drop) and steady-state (terminal velocity) methods; steady-state analysis reduced uncertainty by $\sim 3\times$.
- Applied regression (LINEST), uncertainty propagation, and experimental error analysis across all studies.



UB SEDS – Propulsion Team

(formulation of SRM fuel and test nozzles)

Objective

- Design, fabricate, and test custom solid rocket motors using in-house developed HTPB-based propellants for the IREC competition.
- Improve nozzle resilience and performance through precise graphite machining and optimized sealing under high-stress firing conditions.

Chemical	Desired	Actual
HTPB	23.04g	23.09
IDP	23.04g	23.6
Oil	5.76g	6.5
Al	14.40g	14.4
CuO	0.504g	0.6
SrCO ₃	0.936g	1.17
AP _{low}	96.85g	96.8
AP _{high}	117.70g	119.2
MDI	5.76g	

Methods & Results

- **Propellant Development:**
 - Mixed and cured a custom solid fuel blend using HTPB and IDP for improved specific impulse and mechanical stability.
- **Graphite Nozzle Fabrication:**
 - Machined nozzles on a precision lathe to within ± 0.01 mm of design tolerance.
 - Cut retention grooves ("channel rings") into the nozzle's outer wall for O-ring insertion, enhancing hoop stress containment during combustion.
- **Structural Sealing:**
 - Implemented rubber O-rings to reinforce the seal between nozzle and casing under explosive internal pressures.
- **Testing:**
 - Conducted static fire tests with inverted motor mounting on a load cell to capture thrust-to-mass ratios.
 - Tests were successfully completed, validating both the custom propellant and nozzle design.
 - Scaled up variants were used successfully on the IREC Competition rocket.



Propulsion Project

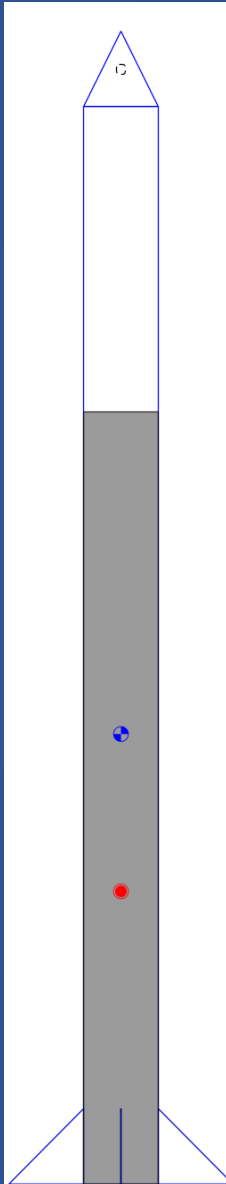
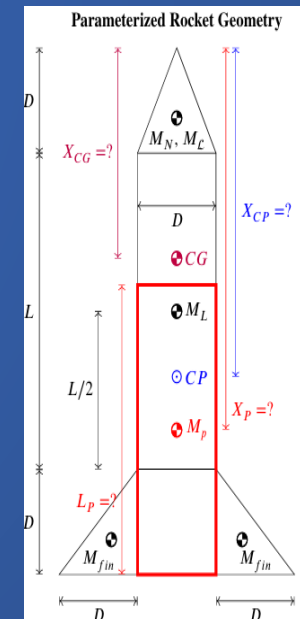
(Optimization and Design of a Single Stage Rocket)

Objective

- Design and optimize a single-stage sounding rocket to maximize payload ratio (λ) under specified mission constraints.
- Study the impact of changing Payload Mass (M_L), Maximum Altitude (h_{max}), and Normalized Acceleration (a_{max}) while keeping structural and aerodynamic parameters constant.
- Develop a MATLAB simulation to generate and evaluate all feasible geometric configurations (diameter D and length L) for each mission case.
- Verify the analytical model using OpenRocket, a 6-DOF rocket simulation software, to ensure agreement with theoretical predictions..

Methods & Results

- **Implemented a MATLAB script to:**
 - Compute mass ratio (R_{opt}), burn time (t_b), and equivalent velocity (W_{eq}) using trajectory theory.
 - Solve for stagnation pressure and temperature assuming ideal expansion.
 - Iterate across **25 million geometry combinations** to extract the best-performing rocket size per case.
- **For each configuration:**
 - Calculated structural mass, propellant volume, and total mass.
 - Used fin-body aerodynamics to determine Center of Pressure (X_{CP}) and compared it against Center of Gravity (X_{CG}) for static margin validation.
 - Maximized payload ratio (λ) while enforcing structural integrity and stability constraints.
- Verified predicted CG/CP locations using OpenRocket—matched within 1–2% of MATLAB outputs.
- **Conducted seven simulation cases:**
 - Varying one mission parameter at a time, while holding the others at “medium” values.
 - Demonstrated how increased payload or altitude requirement affects rocket sizing, propellant need, and structural ratio (ϵ).



FEA & CFD Project

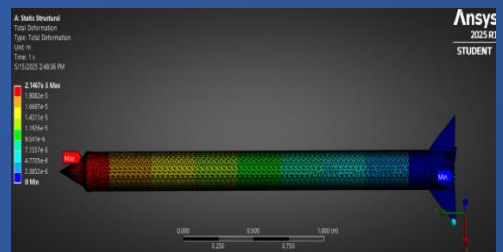
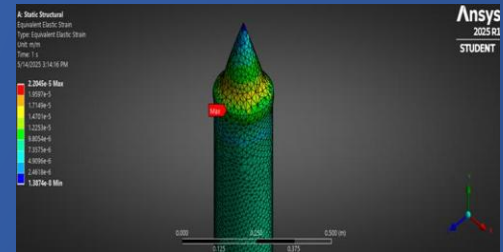
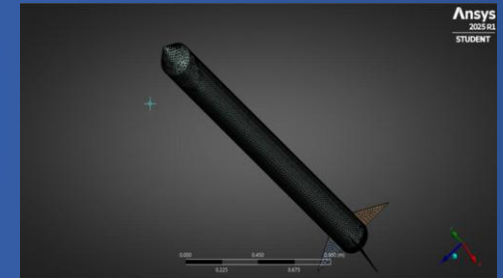
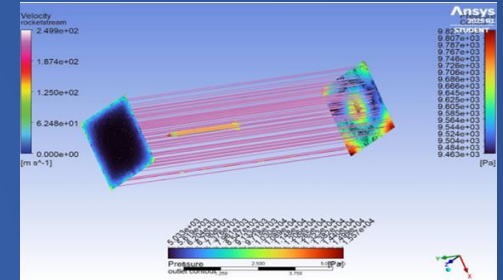
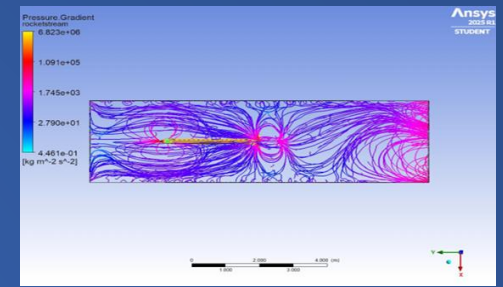
(Finite Element Method Using ANSYS & MATLAB)

Objective

- Assess the structural viability of a student-designed sounding rocket using CFD and FEA tools.
- Analyze aerodynamic loading and resulting deformation, stress, and strain to evaluate performance across materials (Aluminum, Titanium, Carbon Fiber)

Methods & Results

- **Design Origin:** Based on a MATLAB-optimized propulsion project (MAE423); CAD modeled in SolidWorks.
- **CFD Analysis (ANSYS Fluent):**
 - Rocket flown at 236 m/s in OpenRocket simulation; speed used for Fluent boundary conditions.
 - Created a wind tunnel enclosure and simulated compressible airflow.
 - **Result:** Max pressure of ~21.5 kPa on the nosecone.
- **Structural FEA (ANSYS Static Structural):**
 - Hollow model (thickness increased to 0.2 mm for meshing) analyzed under CFD-derived pressure.
 - Fixed base and pressure applied to nosecone.
 - Compared total deformation, stress, and strain for three materials:
 - **Aluminum Alloy:** ~21.5 μm max deformation, moderate stress levels.
 - **Titanium Alloy:** 26% less deformation than aluminum; higher strength but much heavier.
 - **Carbon Fiber:** Over 14 $\times$ more deformation than titanium; unsuitable under compressive loads.
- **Verification:** Mesh sensitivity analysis confirmed result consistency (<25% variation).



Spacecraft Project

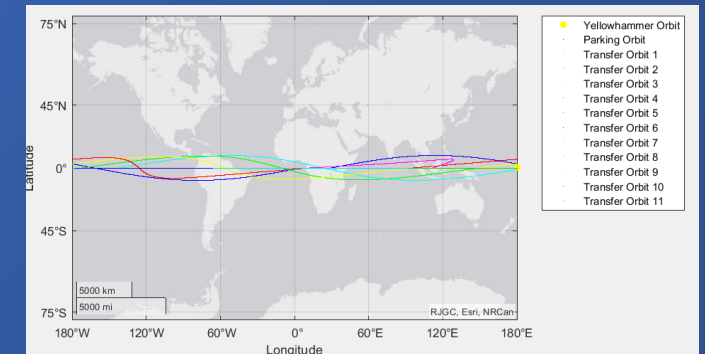
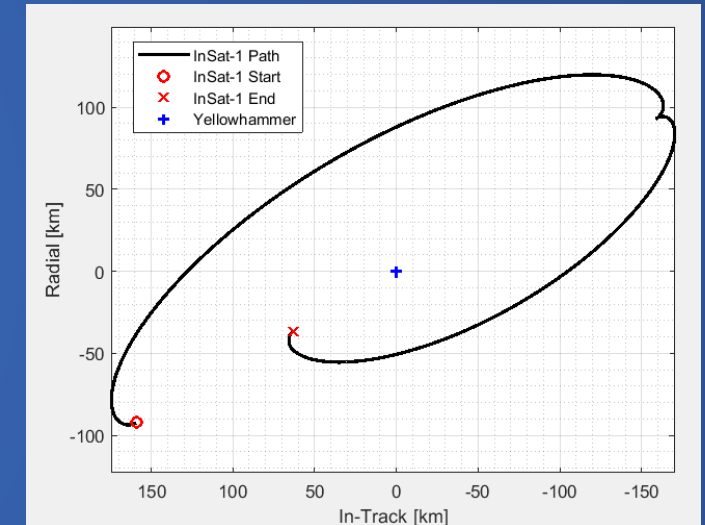
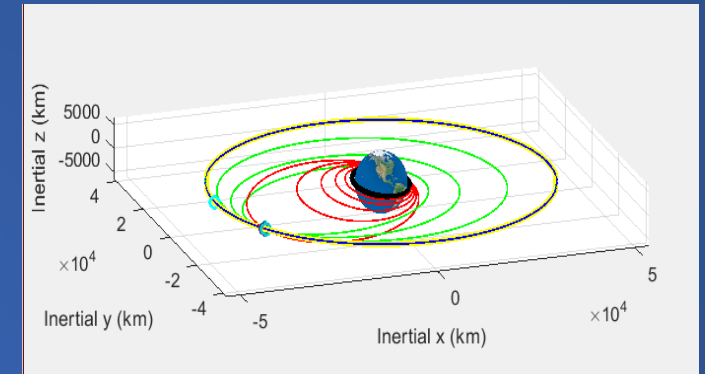
(Fulfilling Space Mission Objectives via Orbital Mechanics & MATLAB simulations)

Objective

- Design and simulate a complete mission for InSat-1 to inspect the geostationary satellite Yellowhammer.
- Demonstrate orbital maneuvering capabilities under propulsion constraints.
- Use onboard sensors to assess satellite health and deliver high-resolution imaging.
- Minimize total ΔV expenditure while ensuring precise phasing and circumnavigation of the target satellite.

Methods & Results

- **Mission Simulation in MATLAB:** Entire mission was designed, simulated, and visualized using custom MATLAB scripts.
- **Propellant-Constrained Maneuver Planning:** Calculated 12 individual burns considering VACCO AFRL Cubesat propulsion limits (max $\Delta V = 468$ m/s per burn)
- **Orbit Transfer & Phasing:** Employed sequential prograde, normal, and retrograde maneuvers to achieve transfer from LEO to GEO and precise relative phasing with Yellowhammer.
- **Timing Optimization:** Waited seven initial orbits and timed burns precisely to align InSat-1 exactly 0.25° behind the target satellite without needing extra station-keeping maneuvers.
- Generated relative motion plots in RIC coordinates using MATLAB.
- Produced ground trace plots for both InSat-1 and Yellowhammer over the mission duration.



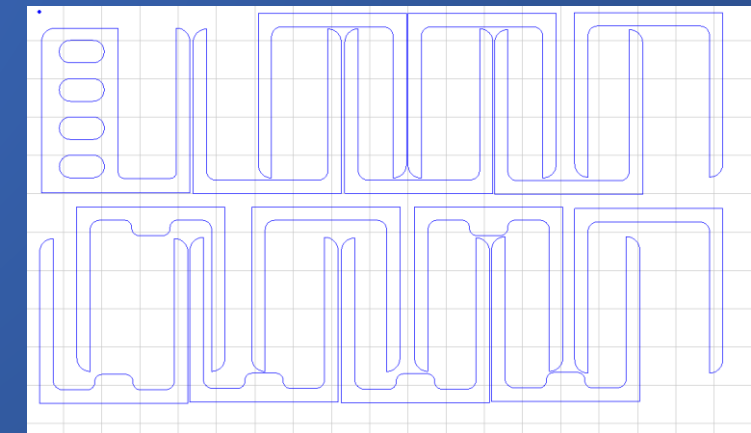
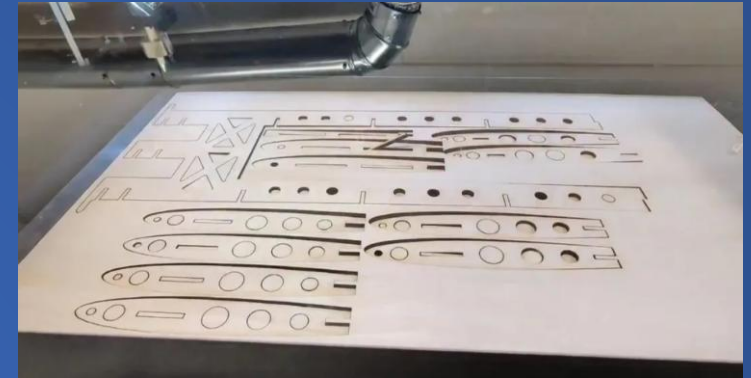
CNC Laser Cutting Specialist

Leadership/Mentorship

- Led laser cutting workshops to train new AIAA DBF team members on safe and effective operation.
- Created a comprehensive instruction and safety manual to onboard new users independently.
- Acted as point-of-contact for troubleshooting, guiding peers through calibration, repair, and usage issues.

Accomplishments & Results

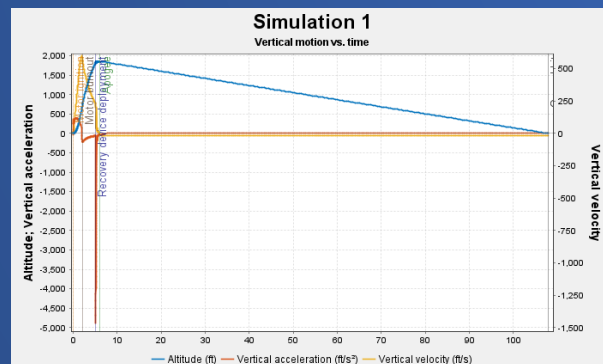
- Reduced cutting tolerances from 0.2 mm to 0.04 mm through precise mirror alignment and focal tuning.
- Maintained peak machine performance during DBF and regional conferences, preventing downtime.
- Designed and engraved custom award plates for official AIAA trophies, contributing to event professionalism.
- Manufactured critical structural components such as fuselage panels, hatches, motor mounts, and ribs with high precision and tight tolerances.



Personal Project – High Powered Rocketry (Construction of LOC IV Rocketry Kit to acquire Level 1 Rocketry License)

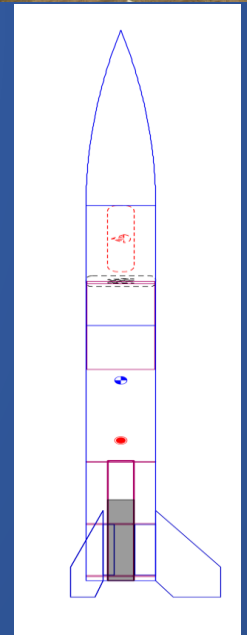
Objective

- Build and modify the LOC Precision LOC IV rocket to meet the requirements for a Level 1 High Power Rocketry Certification.
- Ensure structural integrity, aerodynamic performance, and flight stability through hands-on fabrication, simulation, and component integration.



Methods & Results

- **Kit Preparation & Finishing:**
 - Manually sanded and smoothed the nosecone to reduce drag and improve aerodynamic efficiency.
- **Hardware Integration:**
 - Assembled and installed motor mount, bulkheads, shock cord, fins, rail guides, parachute system, and charge bay with high precision.
 - Modified internal structures for improved load distribution and recovery reliability.
- **Motor Selection & Simulation:**
 - Selected a certified Level 1 solid motor based on Tripoli requirements.
 - Ran performance simulations in OpenRocket, incorporating non-ideal launch conditions to validate motor selection and trajectory.
- **Pre-Flight Status:**
 - Rocket assembly and simulations confirmed readiness for certification flight.
 - Actively monitoring weather and scheduling a launch window to complete the Level 1 certification.



Undergraduate Teaching Assistant (MAE 385: Eng. Materials Lab)

Leadership/Mentorship

- Guided students one-on-one during lab sessions, ensuring they followed procedures and understood concepts.
- Acted as a resource for questions during experiments, supporting learning beyond lectures.
- Enforced lab safety protocols and proactively educated students on best practices.

Accomplishments & Results

- Independently managed grading for all lab quizzes, maintaining fairness and consistency.
- Ensured experimental accuracy by verifying student data, reducing reporting errors.
- Contributed to a safer and more educational lab environment by monitoring and mentoring 20+ students per session.



Thank You for viewing my Engineering Portfolio!



*If you have any questions or concerns,
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